From a Single Axiom to Four-Dimensional Spacetime: A Readout Mechanism for Discrete Ontology

Abstract

This work establishes a discrete ontological framework whose sole starting point is a single axiom: *zero never stays still*. No manifold, metric, probability, or action is introduced. Words, graphs, divergence identities, zero-sum conservation, and integer multiplicities are all derived structures. We establish a layered index system distinguishing the evolution layer from the historical layer, and give a strict parameter account: one axiom, one free parameter, two named inputs. On this basis we present the main derivation chain: zero-sum constraint $\to$ space of balanced words $\to$ self-identifying coarse-graining partition $\to$ spectral dimension readout. At $n = 16$, the participation index of the height-level partition is $q = \sum_b \omega_b^2 = 0.55777859$, which lies in the interval $(1/2, 3/5)$; the functional $F_D = \binom{D}{2} q^D$ then has a unique finite maximum at $D = 4$. We explicitly list all open items and conditional boundaries: the necessity of the quantum contextuality threshold is unproven; the partition criterion has been shown non-unique; and $D = 4$ holds only for $n \in \{16, 20\}$.

Keywords: discrete ontology; zero-sum constraint; coarse-graining; spacetime dimension; spectral dimension; KCBS inequality

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1 Introduction

The spacetime dimension $D = 4$ is an input rather than a conclusion within the standard physical framework [1,2]. The axiomatic systems of quantum mechanics and general relativity likewise lack a common derivation basis. Our approach is to start from a discrete substrate containing no geometry, no probability, and no action, such that dimension becomes a computable readout quantity rather than an intrinsic property of microscopic degrees of freedom.

The core stance is: dimension in discrete systems is inherently a readout quantity at the coarse-graining scale, not a microscopic degree of freedom. The framework therefore does not posit a "microscopic dimension"; it only posits a "readout dimension." The method proceeds in four steps: axiom $\to$ combinatorial structure $\to$ partition $\to$ readout.

The paper is organized as follows: §2 states the single axiom and its mathematical formulation; §3 derives words, graphs, the divergence dictionary, and zero-sum conservation; §4 establishes layered indices; §5 gives the object dictionary; §6 registers the parameter account; §7 presents the main derivation chain from balanced words to dimension readout; §8 summarizes proven results; §9 lists conditions and open items; §10 is discussion.

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2 The Axiom and Its Mathematical Formulation

2.1 The Single Axiom

Axiom 1 (Zero Never Stays Still). The equilibrium (closed) state is not stationary; it moves continuously. Motion does not terminate at any point in time. Motion carries no weight—there is no selection rule in the system.

2.2 Mathematical Formulation

The three postulates of the axiom correspond to the following structures:

$$Z_0 = \underbrace{(\{\pm 1\},\, \iota)}_{\text{sign structure}} + \underbrace{(\mathcal{W}_n,\, \mu)}_{\text{closure constraint + counting measure}}. \tag{2.1}$$

Sign structure. Motion has two directions, giving $\{\pm 1\}$. Zero is not a stationary point but an equilibrium relation, and is therefore expressed as the quotient by an involution $\iota : \{\pm 1\} \to \{\pm 1\}$ ($\iota^2 = \mathrm{id}$, $\iota(x) \neq x$):

$$\text{zero} = \{\pm 1\}/\iota. \tag{2.2}$$

Closure constraint. Motion along a graph $\Gamma = (C, E)$ gives a divergence $d_w(v) = \#\{\text{entering } v\} - \#\{\text{leaving } v\}$ satisfying the zero-sum identity

$$\sum_{v \in C} d_w(v) \equiv 0, \qquad \text{closed} \iff d_w = 0. \tag{2.3}$$

Counting measure. No weight data exist in the system, so no motion can be singled out—every allowed motion is instantiated (full branching), with multiplicity given by counting. The only available measure on $\mathcal{W}_n$ is the uniform counting measure

$$\mu(A) = \frac{|A|}{|\mathcal{W}_n|}, \qquad |\mathcal{W}_n| = \binom{n}{n/2}. \tag{2.4}$$

Proposition 2.1 (Uniqueness of the measure). The uniform counting measure (2.4) is not an assumption.

Proof. Let μ' be any measure on $\mathcal{W}_n$. If $\mu' \neq \mu$, then there exist A, B such that $\mu'(A)/|A| \neq \mu'(B)/|B|$, i.e., there is a bias toward some motions. By the third postulate of the axiom (no selection rule), μ' is incompatible with the axiom. ■

Remark. The third postulate is a minimality argument: it establishes that no weight data exist within a single-axiom system, rather than asserting that probability is physically impossible. If probability were allowed, the full-branching conclusion would fail.

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3 Derived Structure

3.1 Words, Graphs, and the Divergence Dictionary

Motion = one move along a directed edge of the graph $\Gamma = (C, E)$; the sequence of steps forms a word $w = (e_1, \dots, e_n)$.

Lemma 3.1 (Update rule). Upon appending an edge $u \to v$, the divergence updates as $d \mapsto d + e_v - e_u$.

Lemma 3.2 (Zero-sum identity). For any word w, $\sum_{v \in C} d_w(v) \equiv 0$.

Proof. Each step simultaneously "enters one place and leaves another," contributing $+1 - 1 = 0$; induction on the number of steps gives the result. ■

Lemma 3.3 (Reachable set). All reachable divergences form the zero-sum hyperplane

$$\{d_w\} = \Big\{ x \in \mathbb{Z}^C : \sum_v x_v = 0 \Big\} = \operatorname{im} B =: H_Q, \qquad \dim H_Q = |C| - 1, \tag{3.1}$$

where B is the incidence matrix of the graph.

Proof. By Lemma 3.2, $\{d_w\} \subseteq H_Q$. Conversely, when Γ is connected, the image of the incidence matrix B spans H_Q, and B is totally unimodular, giving integer reachability [3]. ■

Remark. Lemmas 3.1–3.3 reduce the zero-sum constraint from an axiom to a theorem.

3.2 Full Branching and Integer Multiplicity

By the no-selection-rule postulate: no weights, probabilities, or choices exist ⇒ every allowed motion is instantiated. Multiplicity is given by counting, hence is integer. This is the watershed between our framework and "borrowed-route" approaches: the latter list full-branching generative semantics as an input; under our axiom it is a corollary.

3.3 Terminal States and Reseeding

Proposition 3.4 (Necessity of reseeding). If the active layer cannot remain empty forever (second postulate of the axiom), then finite-lifetime models must be accompanied by reseeding.

Proof. A four-model comparison shows: finite-lifetime models without reseeding necessarily extinguish; periodic-clearance plus reseeding models persist. ■

Proposition 3.5 (Geometric necessity of terminal states). Requiring a stable spectral dimension forces loop-length truncation; branches that do not close enter a terminal state at their lifetime.

Proof. In the closed-loop graph (nodes = rotation classes of zero-sum cyclic words of period T), without truncation the average degree grows unbounded with T (3.60 at $T = 8$, 9.50 at $T = 18$), leading to non-manifold discretization and no stable spectral dimension. ■

Remark. This is the only conditional derivation in the framework: if the geometric requirement is abandoned, the terminal postulate loses its justification.

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4 Layered Indices

The physical world resides in the evolution layer. The same quantity manifests differently at different layers; layers must be aligned before comparison.

Definition 4.1 (Four-layer index).

- L0: axiom layer—postulates, integer multiplicities, local compensatory moves;
- L1: exact words (historical layer)—only increasing, irreversible;
- L1′: global closed-class layer—recording and readout;
- L2: evolution layer (active layer)—has its own update law.

Proposition 4.2 (Stratification Precedence). When two conclusions conflict, the default explanation is not contradiction but different layers; only opposites at the same layer constitute a genuine contradiction.

Reasoning. The layer index is part of the assertion; an assertion omitting the layer index is incomplete. ■

Corollary 4.3. The "readout" (π) does not occupy a layer index—it is a named input (§6).

Reasoning. If inputs were treated as layers, any conflict between two conclusions could be dissolved by "different layers," which is the opposite of the self-consistency requirement.

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5 Object Dictionary

5.1 Kinematic Objects

Symbol	Name	Definition
$\Gamma = (C, E)$	graph	positions and steps; finite, connected
$w = (w_1, \dots, w_n)$	word	sequence of steps
d_w	divergence	entering count − leaving count
H_Q	zero-sum hyperplane	$\{x : \sum x_v = 0\}$
$\mathcal{W}_n$	balanced word space	$\{w \in \{\pm 1\}^n : \sum w_k = 0\}$
s_k	prefix sum	$\sum_{i \le k} w_i$, $s_0 = s_n = 0$

5.2 Intrinsic Quantities

$$\text{height} : h(w) := \max_{0 \le k \le n} |s_k|, \qquad \text{area} : a(w) := \sum_{k=1}^{n} s_k. \tag{5.1}$$

Proposition 5.1. Both h and a are anchored at $s_0 = 0$ (i.e., zero), not at n.

Proof. Both are given directly by the prefix-sum sequence, which starts at $s_0 = 0$. ■

5.3 Readout Objects

Symbol	Name	Definition
π	partition (readout)	$\pi : \mathcal{W}_n \to \{1, \dots, k\}$
$\mathcal{B}_b$	block	$\{w : \pi(w) = b\}$
ω_b	block weight	$\lvert\mathcal{B}_b\rvert / \lvert\mathcal{W}_n\rvert$
q	participation index	$\sum_b \omega_b^2$
F_D	dimension-selection functional	$\binom{D}{2} q^D$
λ	normalized spectrum	top three block weights, descending and normalized

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6 Parameter Account

$$Z_0 + \underbrace{n}_{\text{free parameter}} + \underbrace{\pi}_{\text{structural input}} + \underbrace{\kappa}_{\text{scale input}} = \text{complete framework.} \tag{6.1}$$

That is: one axiom, one free parameter, two named inputs.

Item	Type	What it buys
Z_0 (three postulates)	axiom	words/graph/steps; closure; full branching; integer multiplicities
n	free parameter	terminal scale; entry point for blocking
π	named input	dimension-selection peak; contextuality threshold; module flow type
κ	named input	scale (1 step = how many physical units)
$\beta\varepsilon$	parameter	effective ladder temperature (height cost), affects state layer only
Γ	named input	topological carrier; specialized to path graph $\mathbb{Z}$ in this work

Proven non-derivable items: absolute scale, D itself (all metric-side structures are graph-independent, so low dimensions cannot be excluded), measure normalization, absolute amplitude.

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7 Main Derivation Chain

7.1 Kinematic Starting Point

Specializing to the path graph:

$$\Gamma = \text{path graph } 0, \pm1, \pm2, \dots; \quad \text{step} = \pm1; \quad \text{closed} \iff s_n = 0. \tag{7.1}$$

$$\mathcal{W}_n = \Big\{ w \in \{\pm1\}^n : \sum_{k=1}^{n} w_k = 0 \Big\}, \qquad |\mathcal{W}_n| = \binom{n}{n/2}. \tag{7.2}$$

Remark. $\mathcal{W}_n$ is a set selected by the zero-sum constraint—no time, no probability, no space.

7.2 Partition and Self-Identification Criteria

Partition $\pi : \mathcal{W}_n \to \{1, \dots, k\}$, block weights $\omega_b = |\mathcal{B}_b|/|\mathcal{W}_n|$.

Definition 7.1 (Self-identification criterion). Partition labels must be recoverable from the block itself (after closure the system has no memory; labels cannot be carried by memory). This requires three structural criteria:

1. **Locally Computable**: the primitive $h(w)$ can be computed from a single word;
2. **Anchored at zero**: $h(w)$ is referenced to $s_0 = 0$, not to n;
3. **Non-Vanishing Bulk**: $\max_b \omega_b$ does not decay with n.

Proposition 7.2. The family of candidate partitions satisfying these three criteria includes but is not limited to: height-level partition (hmax), longest-run partition (maxrun), first-flip partition (firstflip), and flip-count partition (nflip).

Proof. Direct verification from the criteria. ■

Remark. Criteria 1 and 2 constrain the single-word primitive $h(w)$; the merging threshold $t^* = \arg\max_h b_h$ (Eq. 7.6) is an ensemble-level coarse-graining rule that depends on the global mode and n, not on individual words. The two are at different layers and do not conflict.

7.3 Height Blocks and the Reflection Principle

$$b_h = \big|\{w \in \mathcal{W}_n : h(w) = h\}\big| = N_{\le}(n, h) - N_{\le}(n, h-1). \tag{7.3}$$

The reflection principle under two-sided absorbing boundaries $\pm(h+1)$ gives [4]:

$$N_{\le}(n, h) = \sum_{m \in \mathbb{Z}} (-1)^m \binom{n}{n/2 + m(h+1)}. \tag{7.4}$$

Explicit computation for $n = 16$:

h	1	2	3	4	5	6	7	8
b_h	256	4118	4858	2518	880	208	30	2

$$\sum_h b_h = 12870 = \binom{16}{8} \quad \checkmark \tag{7.5}$$

Merging rule (taking the modal level as threshold):

$$t^* := \arg\max_h b_h; \quad h > t^* \text{ form independent blocks}; \quad h \le t^* \text{ merge into one block.} \tag{7.6}$$

For $n = 16$, $t^* = 3$, and the unnormalized block sizes are

$$[|\mathcal{B}_b|] = [9232,\ 2518,\ 880,\ 208,\ 30,\ 2], \qquad k = 6. \tag{7.7}$$

7.4 Three Quantum Gates

Let $\lambda = (\lambda_1, \lambda_2, \lambda_3)$ be the top three block weights in descending order, normalized, and $q = \sum_b \omega_b^2$.

Definition 7.3 (Three gates).

- **Gate 1 (survival window)**: $q \in (1/2,\ 3/5)$;
- **Gate 2 (contextuality)**: $S_{\max} = \mu_1\lambda_1 + \mu_2(\lambda_2 + \lambda_3) > 2$, where $\mu_1 = \sqrt{5}$, $\mu_2 = (5 - \sqrt{5})/2$;
- **Gate 3 (density)**: rank{prime exponent difference vectors} $= k - 1$.

The μ coefficients in Gate 2 are the von Neumann trace inequality coefficients of the KCBS inequality [5,6]; the non-contextual bound is 2.

7.5 Dimension-Selection Functional and Peak Condition

$$F_D = \binom{D}{2} q^D. \tag{7.8}$$

Here $\binom{D}{2} = \dim \mathfrak{so}(D-1, 1)$ (the dimension of the Lorentz algebra, i.e. the number of independent Lorentz generators), and q^D is the cost of requiring simultaneous record across D directions.

Lemma 7.4 (Peak condition).

$$\frac{F_D}{F_{D-1}} = \frac{D}{D-2} q \quad \Longrightarrow \quad \text{unique finite maximum at } D = 4 \iff q \in (1/2,\ 3/5). \tag{7.9}$$

Proof. $F_D/F_{D-1} > 1 \iff q > (D-2)/D$. At $D = 4$ this requires $q > 1/2$; at $D = 5$, $q < 3/5$. Thus $q \in (1/2, 3/5)$ simultaneously gives $F_4 > F_3$ and $F_5 < F_4$; for $D \ge 5$, $(D-2)/D \ge 3/5 > q$, so F is strictly decreasing for $D \ge 5$. At the boundary: $F_0 = F_1 = 0$, $F_2 = q^2$, $F_3 = 3q^3$, and when $q > 1/2$, $F_3/F_2 = 3q > 3/2 > 1$, so $D \le 3$ does not exceed F_4. The maximum is unique. ■

7.6 Layer Distinction

The three gates read different layers: the blocking layer (k, ω_b, q) is determined by pure counting and does not depend on $\beta\varepsilon$; the state layer (spectrum of ρ, $S_{\max}$) is determined by the measure $\omega \propto m\, e^{-\beta\varepsilon t}$ and varies monotonically with $\beta\varepsilon$.

Consistency check. At $n = 16$, $k = 6$ for all $\beta\varepsilon$; $S_{\max}$ increases monotonically from 2.006279 ($\beta\varepsilon = 0$) to 2.235986 ($\beta\varepsilon = 2$).

7.7 7.7 Explicit Solution for $n = 16$

$$q = \frac{9232^2 + 2518^2 + 880^2 + 208^2 + 30^2 + 2^2}{12870^2} = \mathbf{0.55777859} \in (1/2,\, 3/5) \quad \checkmark \tag{7.10}$$

$$(\lambda_1, \lambda_2, \lambda_3) = (0.73095804,\ 0.19936659,\ 0.06967538), \tag{7.11}$$

$$S_{\max} = \sqrt{5}\,(0.73095804) + \frac{5-\sqrt{5}}{2}(0.19936659 + 0.06967538) = \mathbf{2.00627871} > 2 \quad \checkmark \tag{7.12}$$

$$\text{rank} = 5 = k - 1 \quad \checkmark \tag{7.13}$$

7.8 Full-Chain Census

$$Z_0 \longrightarrow \mathcal{W}_n \longrightarrow \pi \longrightarrow (n, \pi) \longrightarrow D = 4. \tag{7.14}$$

A census over $n = 6, \ldots, 48$ (22 even values) $\times$ 4 self-identification candidate partitions (88 combinations) yields:

$$(n, \pi) \in \{(16, \texttt{hmax}),\ (20, \texttt{maxrun})\} \qquad (2/88 = 2.3\%). \tag{7.15}$$

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8 Summary of Proven Results

1. **One axiom**, three postulates each buying one thing (proven).
2. **Zero-sum constraint is a theorem, not an axiom** (Lemmas 3.1–3.3, proven).
3. **Counting measure is unique** (Proposition 2.1, proven).
4. **Reseeding is forced by the requirement of persistent motion** (Proposition 3.4, proven).
5. **Terminal states are forced by geometric stability** (Proposition 3.5, conditionally derived).
6. **Self-identification criteria restrict the candidate partition family** (Proposition 7.2, proven).
7. **The height-level partition passes all three quantum gates and peaks at $D = 4$** (§7.7, computed).
8. **Peak at $D = 4 \iff q \in (1/2, 3/5)$** (Lemma 7.4, proven).
9. **D cannot be determined from proven structures; absolute scale is non-derivable** (§6, proven non-derivable).
10. **Second route**: the cycle space $Z_1(K_4)$ and displacement space H_Q form an axial–polar vector duality in three dimensions via the Hodge star $\star : \Lambda^2\mathbb{R}^3 \to \mathbb{R}^3$, independently locking $d = 3$ without the F_D peak condition (see companion paper [7]).

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9 Conditions and Open Items

This section is of equal importance to the main text.

#	Condition	Status
1	**Quantumness required** (Gate 2: $S_{\max} > 2$)	Premise. Critical bottleneck of the entire chain. Necessity unproven
2	**Partition criterion non-unique**	Proven non-unique: the three structural criteria are simultaneously satisfied by hmax and firstflip
3	Necessity of the three structural criteria	Unproven (but all are preference-free)
4	Exhaustiveness of the candidate partition family	Unproven (only 4 self-identification candidates checked)
5	**Conditionality of** $D = 4$	Not true for all n: only $n \in \{16, 20\}$ give peak $= 4$. The Hodge duality route independently locks $d = 3$, but requires the three-dimensional geometric realization premise
6	Scan range for n	Only $n = 6, \ldots, 48$
7	Status of κ	Fixed once or reselected each round; indistinguishable within the framework
8	Arrow of time	Still open
9	Continuum limit / metric / field equations	Not addressed in this work; see companion paper [7]
10	**Truncation renormalization in Gate 2**	$S_{\max} > 2$ is obtained by taking the top three blocks and renormalizing (discarding $h = 6, 7, 8$, total 240 words). Without renormalization $S_{\max} = 1.9946 < 2$, failing the contextuality bound. Gate 2 is currently conditionally justified
11	**Gate 2 fixed at** $\beta\varepsilon = 0$	$\beta\varepsilon$ is a free parameter, but the census uniqueness holds at $\beta\varepsilon = 0$ (pure counting); otherwise any partition passing Gates 1 and 3 could have $S_{\max}$ cross 2 by taking $\beta\varepsilon > 0$

#	Condition	Status
12	**Functional form** $F_D = \binom{D}{2} q^D$	Taking the number of Lorentz generators times q^D and choosing the physical dimension by the maximum is a readout functional hypothesis on par with π, not a theorem derived from Z_0

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10 Discussion

$$Z_0 \Longrightarrow \mathcal{W}_n \Longrightarrow \pi \text{ (self-identifying)} \Longrightarrow D = 4.$$

All results are conditional on quantumness; the axiom itself is unchanged.

The framework's free content consists of exactly four items: one axiom Z_0, one free parameter n, two named inputs π and κ.

The two weakest links at present are: link (partition non-uniqueness, proven) and the quantumness premise of link (unproven). Subsequent work targets these two links, as well as establishing the continuum limit and field-theoretic stage.

Claims not made: Z_0 has been modified; the chain holds unconditionally; π has been uniquely determined; $D = 4$ holds for all n; contextuality resolves the arrow of time; this work provides a continuum limit, metric, or gravitational field equations.

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